How Can a Computer Recognize Hand Letters? Image Preprocessing, Augmentation and Hand-Shape Features

Ton Heijligers (852935308) [1] **Youri de Waal (852843968)** [1] **Danny Linden (85261859)** [1]

Abstract

Can a computer identify a letter shown by a hand in a photograph? We study targeted augmentation, a second detection attempt after image preprocessing, and explicit hand-shape descriptions. This standalone working paper develops the feature contribution; the complete group version includes the supplied detection-recovery text. Eight feature combinations are compared on 60,251 accepted images. Own categories do not consistently improve recognition. A small gain from world-joint measurements also depends on the coordinate source and learner. Synthetic orientation improvements do not establish better recognition for new users or a live camera.

1. What do we investigate?

The American Sign Language (ASL) hand alphabet represents letters using hand shapes and movements. We study static A–Z images, not continuous signing or the movements of J and Z.

Our shared question is: *Which changes help a computer locate a hand and recognize its letter?* The group paper uses three research questions:

1. **RQ1 – Augmentation:** Does targeted augmentation reduce confusion more effectively than general augmentation?
2. **RQ2 – Detection recovery:** Can a 3×3 dilation filter recover visually plausible landmarks when MediaPipe initially detects no hand?
3. **RQ3 – Hand shape:** Can own hand-shape categories and joint measurements supplement or replace Shin's landmark directions?

We test these changes separately. Accepting more hand detections does not automatically improve letter classification.

[1]Open Universiteit.

Working version of a joint study project.

2. How does recognition work?

MediaPipe first estimates 21 points around the wrist, knuckles and fingertips (Google AI Edge, n.d.). Think of them as dots on a simple hand skeleton. If a dot is misplaced, the hand description may also be wrong.

We describe the points using distances and directions following Shin et al. (2021). A classifier learns from these descriptions and their letters, then predicts images excluded from its fitting data. We use SVM and LightGBM (LightGBM developers, n.d.); implementation settings are in Appendix A.

3. How do we test?

3.1. Separate fitting and testing

The Kaggle ASL Alphabet dataset supplies 78,000 A–Z training images (Nagaraj, 2018). Exactly one accepted hand was obtained for 60,251 images. Only those images enter the feature comparison, so its scores do not describe the complete recognition pipeline over every source image.

Five fixed folds separate blocks of consecutive file numbers. Each model fits four folds and predicts the remaining fold, until every image has one out-of-fold prediction. All feature variants share test sources. A source and its derivatives stay in the same fold. These blocks are not verified participants.

Macro-precision averages letter precision equally across A–Z. Macro-F1 also accounts for missed true letters. F1 ranges from zero to one; higher is better. Technical definitions and settings are in Appendix A.

TODO [RQ1/RQ2]: Integrate the augmentation contribution and executable detection-recovery implementation. The group version already contains the supplied recovery results.

3.2. RQ3: eight hand descriptions

We compare distances (D), Shin directions (S), own categories such as finger flexion/spreading (E), and joint measurements (J). D is always present; independently including or omitting S/E/J gives eight combinations. Both learners receive the same combinations and test images. Appendix C explains the categories and inspection tool.

E and J use estimated world points; D and S use image points. A separate control derives J from exactly the same points as Shin. This distinguishes coordinate-source changes from a different description of the same points (Appendices A–B).

After fitting, we shuffle a feature group across test images and measure precision/F1 losses (scikit-learn developers, n.d.). Labels and fitted models remain unchanged. These tests use 13,000 fixed held-out images; the eight-way classification comparison uses all 60,251 accepted images.

4. Results

4.1. RQ3: which combination works better?

Table 1 compares the eight descriptions. World-joint measurements describe an estimated 3D hand, rather than independently measured anatomy. Relative to Shin DS, own categories give 22 fewer correct SVM predictions. World-joint measurements give 67 more; both additions together give 3 fewer. For LightGBM, adding categories changes F1 from 0.9678 to 0.9676. Without directions, categories and world joints give 318 more correct predictions than distances alone. Appendix B gives detailed scores, paired changes and the same-coordinate control. The same-coordinate angle control gains only four SVM predictions. The larger world-point gain therefore cannot be attributed to the angle formula alone.

4.2. Which information does the model use?

For SVM with DS and own categories, shuffling thumb ip flexion gives the largest mean F1 loss (0.08 percentage points). For LightGBM, it is middlering spreading (0.03 percentage points). These are dependencies of the fitted models. A small loss does not establish that a feature is useless: other features may supply the same information. See Appendix B. A feature used by a fitted model need not improve a newly fitted model when added. Importance depends on the learner and other available features (scikit-learn developers, n.d.).

4.3. Brief practical explorations

Alongside the main feature comparison, we briefly tried ideas for left/right handling and G/H camera views. Mirror training and hand normalization help on artificial mirrored tests but do not improve original-view classification. For G/H, usual ASL uses an inward or sideways palm (Vicars, n.d.), unlike selected training views. Rotating training landmarks teaches the artificial geometry, but two selected live G examples remain H. These short exploratory follow-ups are not evidence of better recognition for new users. Appendices D–F give the attempted methods, scores and examples.

5. Discussion and provisional conclusion

Own categories do not consistently improve Shin. Under these settings, categories and joint measurements also do not fully replace the directions. The small SVM world-joint gain does not recur with LightGBM, and coordinate source is a relevant alternative explanation. Retraining measures adding/removing information; permutation measures dependence of a fitted model, where related features may substitute for one another.

The brief left/right and G/H explorations show limits of synthetic changes: gains on artificial presentations do not establish real-user improvement. They are practical side checks, not independent validation of the feature comparison.

The folds were inspected during development and are not verified people, so this is exploratory validation, not an untouched final test. A high score here does not guarantee webcam performance for a new person. Scores also differ from Shin's evaluation because source selection and splitting differ.

TODO [TEAM]: Complete the shared interpretation once the augmentation contribution and recovery implementation/visual assessment are supplied.

Text and software were prepared with AI assistance; the authors remain responsible for content. Webcam images require consent and remain local.

References

Google AI Edge. MediaPipe hand landmarker. Official documentation, n.d. URL https://ai.google.dev/edge/mediapipe/solutions/vision/hand_landmarker. Accessed 8 November 2026.

LightGBM developers. LightGBM: Parameters. Official documentation, n.d. URL https://lightgbm.readthedocs.io/en/stable/Parameters.html. Accessed 8 November 2026.

Nagaraj, A. ASL alphabet. Kaggle dataset, 2018.

Table 1. Eight representations on the same 60,251 held-out sources. All contain D. P is macro-precision (%); F1 is macro-F1. Each source is predicted only by a model fitted without it.

	SVM		LightGBM	
Hand description	P (%)	F1	P (%)	F1
Distances + directions	97.39	0.9739	96.80	0.9678
Distances only	94.43	0.9438	93.63	0.9355
DS + own categories	97.36	0.9735	96.78	0.9676
D + own categories	94.64	0.9460	93.72	0.9364
DS + world joints	97.54	0.9752	96.71	0.9669
D + world joints	95.05	0.9501	93.83	0.9377
DS + categories + joints	97.39	0.9738	96.78	0.9675
D + categories + joints	94.97	0.9493	93.87	0.9379

URL https://www.kaggle.com/datasets/grassknoted/asl-alphabet. Accessed 8 November 2026.

scikit-learn developers. Permutation feature importance. Official scikit-learn documentation, n.d. URL https://scikit-learn.org/stable/modules/permutation_importance.html. Accessed 8 November 2026.

Shin, J., Matsuoka, A., Hasan, M. A. M., and Srizon, A. Y. American sign language alphabet recognition by extracting feature from hand pose estimation. *Sensors*, 21(17):5856, 2021. doi: 10.3390/s21175856. URL https://doi.org/10.3390/s21175856.

Vicars, W. ASL feedback: fingerspelled G and H. ASL University / Lifeprint, n.d. URL https://www.lifeprint.com/asl101/topics/feedback.htm. Accessed 8 November 2026.